Capstone Project & Incident Response Report

Web Application Penetration Testing using DVWA and Metasploitable2

1. Introduction

This project focused on learning practical cybersecurity concepts through a controlled penetration testing environment using Kali Linux, DVWA, and Metasploitable2. The project helped in understanding vulnerabilities, exploitation techniques, web security flaws, and basic incident response practices.

2. Project Objectives

- Understand penetration testing methodology.
- Perform vulnerability scanning and service enumeration.
- Test web application vulnerabilities.
- Learn exploitation using Metasploit Framework.
- Study security mitigation techniques.
- Understand incident response concepts.

3. Technologies and Tools Used

The following tools and platforms were used during the project:

- Kali Linux
- DVWA (Damn Vulnerable Web Application)
- Metasploitable2
- Nmap
- Metasploit Framework
- VirtualBox
- Apache2 and MySQL services

4. Lab Environment Setup

A virtual penetration testing lab was created using VirtualBox. Kali Linux was configured as the attacker machine, while DVWA and Metasploitable2 acted as vulnerable targets. Apache2 and MySQL services were configured to run DVWA locally through localhost.

5. Reconnaissance and Scanning

Network reconnaissance was performed using Nmap. Open ports, services, and vulnerabilities were identified on the target machine. Commands such as 'nmap -sV' and 'nmap --script vuln' were used to gather detailed service information and identify potential attack vectors.

6. Web Application Testing

DVWA was used to understand common web vulnerabilities. SQL Injection testing was performed by entering malicious payloads into input fields. This demonstrated how insecure input handling can expose sensitive database information and compromise applications.

7. Exploitation Phase

Metasploit Framework was used to explore publicly known vulnerabilities in the target environment. The VSFTPD backdoor module was tested against the vulnerable machine to understand the exploitation process and how attackers gain unauthorized access.

8. Incident Response Understanding

Basic incident response concepts were studied during the project. This included identifying vulnerable services, understanding attack behavior, analyzing system exposure, and learning methods to reduce risks after detecting vulnerabilities.

9. Security Mitigation Techniques

The following mitigation techniques were studied:

- Regular software updates and patch management.
- Input validation and sanitization.
- Use of strong passwords and access control.
- Disabling unnecessary services.
- Network monitoring and firewall configuration.
- Performing periodic vulnerability assessments.

10. Learning Outcomes

This project provided practical exposure to ethical hacking and cybersecurity workflows. It improved understanding of penetration testing phases including reconnaissance, scanning, exploitation, and mitigation. The project also strengthened knowledge of Linux commands, networking, and web application security.

11. Challenges Faced

During the project, configuration issues related to Apache2, MySQL, and DVWA setup were encountered. Troubleshooting these problems helped in understanding service management, configuration files, and debugging methods in Linux environments.

12. Conclusion

The project successfully demonstrated penetration testing concepts in a safe and controlled environment. It provided valuable hands-on experience with vulnerability assessment, web security testing, and exploitation tools. Overall, the project enhanced practical cybersecurity knowledge and problem-solving skills.

Additional Notes 1

The project emphasized ethical and responsible cybersecurity practices. All testing activities were performed within a controlled virtual lab environment for educational and learning purposes only.

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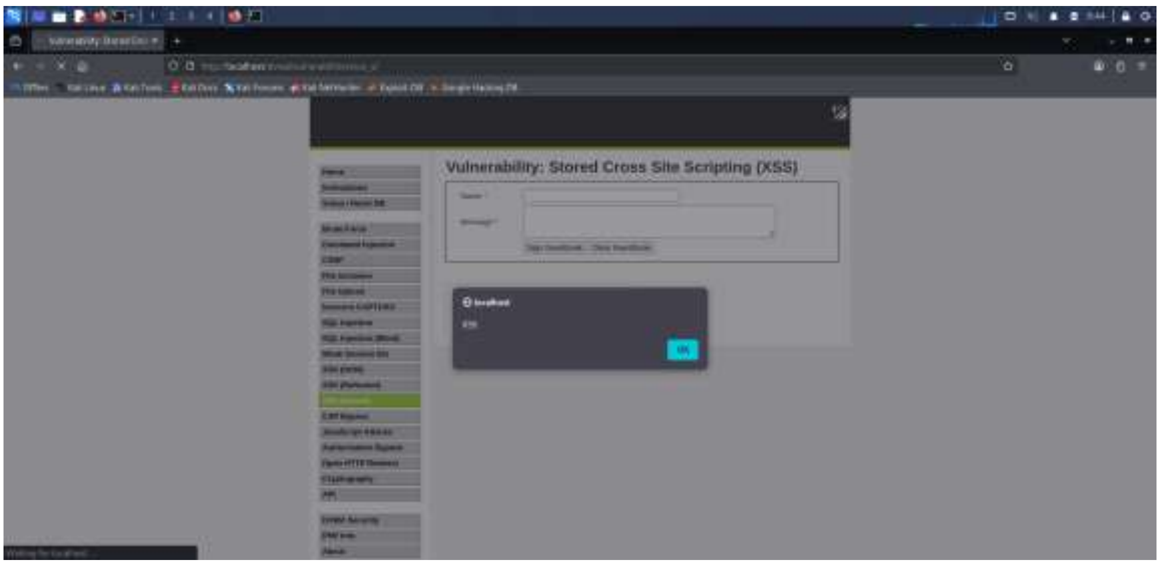
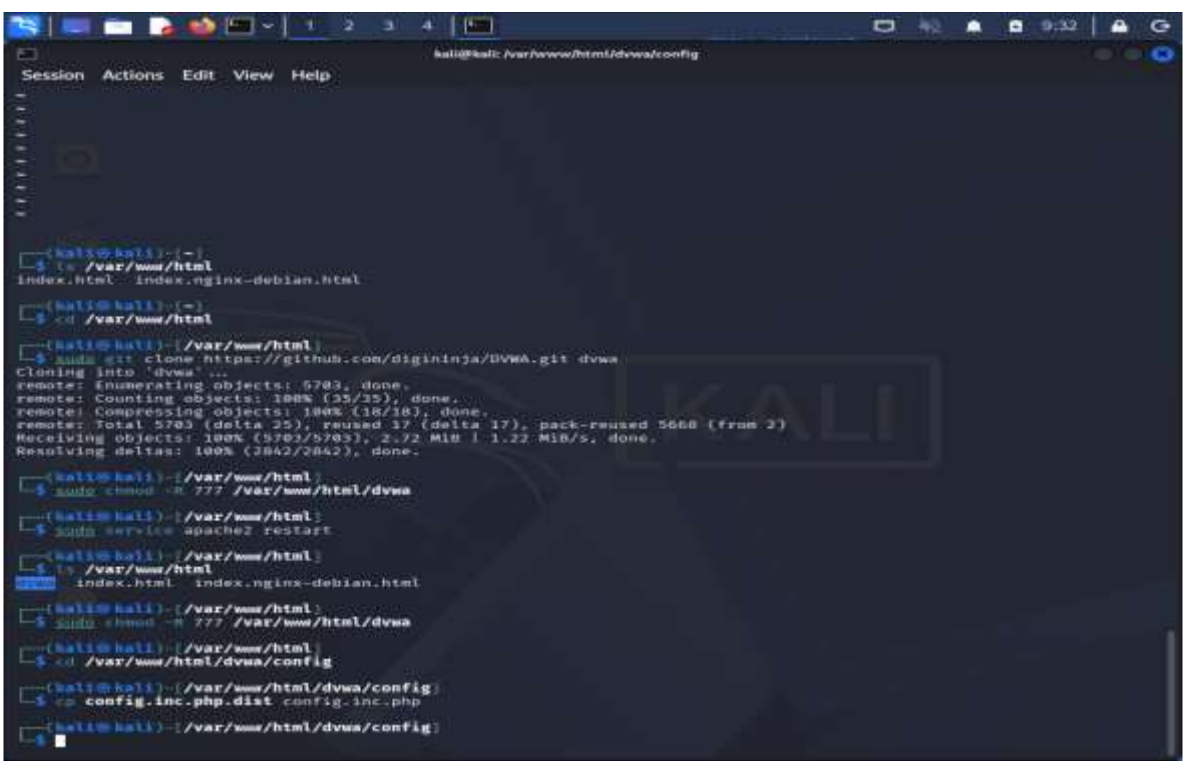

Vulnerability: Stored Cross Site Scripting (XSS)


Session Actions Edit View Help
kali@kali: /var/www/html/dvwa/config
(kali@kali)-[~]
$ ls /var/www/html
index.html index.nginx-debian.html
(kali@kali)-[~]
$ cd /var/www/html
(kali@kali)-[/var/www/html]
$ sudo git clone https://github.com/digininja/DVWA.git dvwa
Cloning into 'dvwa'...
remote: Enumerating objects: 5703, done.
remote: Counting objects: 100% (35/35), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 5703 (delta 25), reused 17 (delta 17), pack-reused 5668 (from 2)
Receiving objects: 100% (5703/5703), 2.72 MiB | 1.22 MiB/s, done.
Resolving deltas: 100% (2842/2842), done.
(kali@kali)-[/var/www/html]
$ sudo chmod -R 777 /var/www/html/dvwa
(kali@kali)-[/var/www/html]
$ sudo service apache2 restart
(kali@kali)-[/var/www/html]
$ ls /var/www/html
index.html index.nginx-debian.html
(kali@kali)-[/var/www/html]
$ sudo chmod -R 777 /var/www/html/dvwa
(kali@kali)-[/var/www/html]
$ cd /var/www/html/dvwa/config
(kali@kali)-[/var/www/html/dvwa/config]
$ cp config.inc.php.dist config.inc.php
(kali@kali)-[/var/www/html/dvwa/config]
$
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